

Name: _____

In the figure at right, D is the midpoint of $\overline{AC}$ and E is the midpoint of $\overline{CB}$.

1. Which side is $\overline{DE}$ parallel to, and how do you know?

2. $AB = 34$. Find DE .

3. $DE = 4x - 3$ and $AB = 6x + 2$. Find x , DE , and AB .

4. $DE = 9.5$. Find AB .

Use coordinates.

5. $\triangle PQR$ has vertices $P(-4, 2)$, $Q(2, 6)$, and $R(4, -2)$. Find the midpoints M of $\overline{PQ}$ and N of $\overline{PR}$.

6. Show that $\overline{MN}$ is parallel to $\overline{QR}$ and half as long.

Answer each question.

7. How many midsegments does a triangle have, and what is each one parallel to?

8. The perimeter of $\triangle ABC$ is 48. Find the perimeter of the triangle formed by its three midsegments.

9. The triangle formed by the three midsegments of $\triangle XYZ$ has perimeter 15. Find the perimeter of $\triangle XYZ$.

10. A trail runs from the midpoint of one leg of a triangular lake to the midpoint of the other. The far shore is 2.4 miles long. How long is the trail?

